

Breast Cancer: Diagnosis to Treatment

Subtypes, Systemic Therapy, Surgery, and Emerging Strategies

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1. Epidemiology & Risk Factors

Breast cancer is the most common malignancy in women worldwide, with approximately 2.3 million new cases annually (GLOBOCAN 2022). Lifetime risk in Western populations is ~12-13%. Major risk factors include age (peak incidence 60-70 years), BRCA1/2 germline variants, dense breast tissue (BI-RADS D), nulliparity, late menopause, HRT use, obesity (post-menopausal), and moderate-high alcohol intake (dose-response relationship).

2. Molecular Subtypes & Treatment Implications

Subtype	ER	PR	HER2	Ki-67	Preferred Systemic Therapy	5-Yr OS
Luminal A	Pos	Pos	Neg	<20%	Endocrine ± CDK4/6i; chemo rarely needed	~92%
Luminal B HER2-	Pos	Low/Neg	Neg	≥20%	Endocrine + CDK4/6i; consider chemo	~83%
Luminal B HER2+	Pos	Any	Pos	Any	Endocrine + anti-HER2 ± chemo	~77%
HER2-enriched	Neg	Neg	Pos	High	Anti-HER2 + chemo (pertuzumab + trastuzumab)	~75%
TNBC	Neg	Neg	Neg	High	Chemo ± immunotherapy (pembro if PD-L1 CPS≥10)	~65%

3. Early-Stage Treatment Paradigms

Neoadjuvant Chemotherapy (NACT) Indications:

- HER2+ tumours ≥ 2 cm or node-positive — enables pathological complete response (pCR) assessment; T-DM1 if residual disease (KATHERINE)
- TNBC — pCR is a surrogate for improved EFS/OS; add capecitabine if residual disease (CREATE-X); olaparib if gBRCA1/2-mut (OlympiA)
- HR+ HER2- — NACT less preferred; consider for inflammatory or T3N+ requiring downstaging

Adjuvant Endocrine Therapy:

Population	Regimen	Duration
Pre-menopausal, low-risk	Tamoxifen	5 years (consider 10 if node+)
Pre-menopausal + OFS (high-risk)	OFS + Exemestane or Letrozole	5 years
Post-menopausal	AI (letrozole/anastrozole/exemestane)	5-10 years
Post-menopausal + CDK4/6i	Abemaciclib + AI (high-risk: ≥ 4 nodes or Ki-67 $\geq 20\%$)	2 years abemaciclib; 5-10 yrs AI

4. Metastatic Breast Cancer — First-Line Principles

- HR+ HER2-: CDK4/6 inhibitor (palbociclib/ribociclib/abemaciclib) + AI (1L); fulvestrant $\pm$ CDK4/6i (2L); PI3K inhibitor (alpelisib for PIK3CA-mut); AKT inhibitor (capivasertib for PIK3CA/AKT/PTEN-alt); sacituzumab govitecan; T-DXd (HER2-low)
- HER2+: Pertuzumab + trastuzumab + docetaxel (1L — CLEOPATRA); T-DXd (2L — DESTINY-Breast03); T-DM1 or lapatinib + capecitabine (3L)
- TNBC: Pembrolizumab + chemo if PD-L1 CPS ≥ 10 (KEYNOTE-522); sacituzumab govitecan; olaparib / talazoparib if gBRCA1/2-mut

CLINICAL PEARL: All newly metastatic patients should undergo repeat biopsy of a metastatic site to confirm receptor status, as discordance from primary occurs in ~20-30% of cases (particularly ER, HER2, and Ki-67).